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MILITARY STANDARD

PRINTED-WIRING AND PRINTED-CIRCUIT TERMS AND DEFINITIONS

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DEPARTMENT OF DEFENSE
WASHINGTON, DC 20301

Printed-Wiring and Printed-Circuit Terms and Definitions

MIL-STD-429C

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense. It shall be used when applicable.

2. Recommended corrections, additions, or deletions should be addressed to the Commander, Rome Air Development Center, ATTN: RBRD, Griffiss AFB, NY 13441.

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FOREWORD

The terms and definitions listed in this standard serve to emphasize the generic nature of the printed-wiring and printed-circuit art. Components, circuits, and assemblies are intimately associated. Processes falling within the art in many cases appear to be disassociated. Some aspects of this art are as follows:

a. Printed wiring and printed circuits may result from entirely different processes, such as photoetching, silk screening, stamping, etc.
b. Printed circuits involve printed patterns of dielectric, magnetic, or semiconductive materials as well as the more common conductive materials.
c. Printed circuits may involve or consist of components both of a three-dimensional and essentially two-dimensional nature.

The terms and their respective definitions are arranged in alphabetical order in section 3 of this standard.

Definitions which indicate a source have been repeated in this standard for convenience. In the event of conflict, the definition in the referenced source takes precedence.

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1. SCOPE

1.1 Purpose. This standard establishes terms and definitions which have specific meaning when applied to printed-wiring and printed-circuit nomenclature. Where two or more acceptable terms are used to designate an item or process, one is defined and the others cross-referenced by the word "see." Where one of two or more items is preferred, the term is defined and the others, included for information only, are cross-referenced to the perferred term by the word "use".

2. REFERENCED DOCUMENTS

2.1 The issues of the following documents in effect on the date of invitation for bids form a part of this standard to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-100 - Engineering Drawing Practices.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

OTHER PUBLICATIONS

DEPARTMENT OF DEFENSE

Cataloging Handbook H6 - Federal Item Name Directory for Supply Cataloging.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

3. TERMS AND DEFINITIONS

3.1 Access holes. Access holes are a series of holes in successive layers, each having a common center or axis. These holes provide access to the surface of the terminal area in one of the layers of a multilayer printed board. (See figure 1.)

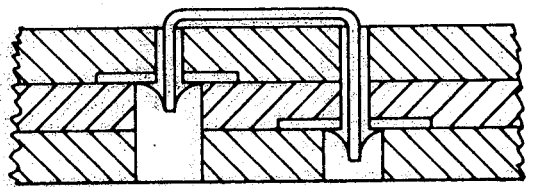


FIGURE 1. Access holes.

3.2 Activating. See "Seeding."

3.3 Additive process. An additive process is a process for obtaining conductor patterns by the selective deposition of conductive material on unclad base material.

3.4 Annular ring. An annular ring is a strip of conductive material completely surrounding a hole.

3.5 Arc resistance. Arc resistance is the resistance of a material to the effects of an arc passing across its surface stated as a measure of the total elapsed time taken to form a conducting path (of material carbonizing by the arc flame) across the surface under prescribed conditions of application of a high voltage, low current arc (as across an insulator).

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3.6 Artwork. Artwork is an accurately scaled configuration which is used to produce the master pattern.

3.7 Artwork master. An artwork master is an accurately scaled configuration used to produce the production master.

3.8 Assembled board. See "Printed-Circuit Assembly" or "Printed-Wiring Assembly."

3.9 Assembly. An assembly is a number of parts or subassemblies or any combination thereof joined together to perform a specific function. (Source: MIL-STD-100)

3.10 Assembly drawing. An assembly drawing is a type of engineering drawing which depicts the assembled relationship of (a) two or more parts, (b) a combination of parts and subordinate assemblies, or (c) a group of assemblies required to form an assembly of higher order. (Source: MIL-STD-100)

3.11 Automated component insertion. Automated component insertion is the act or operation of assembling discrete components to printed boards by means of computer-controlled component-insertion equipment.

3.12 B-Stage. B-Stage is an intermediate stage in the reaction of a thermosetting resin in which the material softens when heated and swells in contact with certain liquids, but does not entirely fuse or dissolve.

3.13 Backpanel. A backpanel is a panel having connector receptacles on one side and terminals (such as solderless wrapped connections) on the other side; used to provide point-to-point electrical interconnections. The point-to-point electrical connections may be printed wiring. (See also "Mother board.")

3.14 Back plane. See "Mother board."

3.15 Base. The base is the insulating support for the printed board.

3.16 Base material. The base material is the insulating material upon which the conductor pattern may be formed. The base material may be rigid or flexible. It may be a dielectric sheet or insulated metal sheet.

3.17 Base material thickness. Base material thickness is the thickness of the base material, excluding metal cladding or conductor thickness.

3.18 Basic dimension. A basic dimension is a theoretically exact dimension without tolerance.

3.19 Basic grid. A basic grid is a theoretically exact, two dimensional network consisting of a set of equally spaced lines, superimposed upon another set of equally spaced parallel lines, such that the lines on one set are perpendicular to the lines of the other, thereby forming square areas. The intersections of the lines provide the basis for a theoretically exact incremental location system.

3.20 Basis material. Basis material is material upon which coatings are deposited.

3.21 Basis metal. A basis metal is a metal upon which coatings are deposited.

3.22 Blank. A blank is an unprocessed piece of base material or metal-clad material having the rough dimensions of a printed board.

3.23 Blister. A blister is an imperfection on the surface of a material caused by a pocket of air or gas beneath the surface.

3.24 Blister, foil. A foil blister is a localized separation between the base material and the foil.

3.25 Blister, interlaminar. An interlaminar blister is a void produced under the surface of a laminated material and is a form of delamination.

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3.26 Board. See "Printed board" and "Multilayer printed board."

3.27 Board thickness. Board thickness is the overall thickness of the base material and all conductive materials deposited thereon.

3.28 Bonding layer. The bonding layer is the adhesive layer (usually B-stage) used to bond individual layers into a laminate.

3.29 Bond strength. Bond strength is a measure, in pounds per square inch, of the force required to separate a layer of foil from the adjoining base by a perpendicular pull applied to the surface of the foil.

3.30 Boss. Use "Terminal area."

3.31 Breakout. Breakout is a condition in which a terminal area and a hole are not concentric, and the hole is not completely surrounded by foil. (See figure 2.)

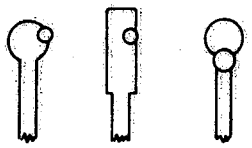


FIGURE 2. Breakout.

3.32 Bridging, electrical. Electrical bridging is the formation of a conductive path or a potential conductive path between conductors.

3.33 Center-to-center spacing. Center-to-center spacing is the distance between the centers of adjacent conductors on any single conductor layer of a printed board. (See figure 3.)

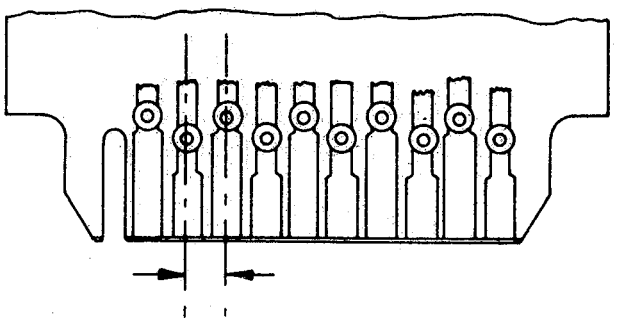


FIGURE 3. Center-to-center spacing.

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3.34 Chemical hole cleaning. Chemical hole cleaning is the chemical process of removing only resin from conductive surfaces exposed in the inside of a hole.

3.35 Circuit. A circuit is the interconnection of a number of electrical devices in one or more closed paths to perform a desired electrical or electronic function.

3.36 Circuit card. See "Printed board."

3.37 Circuit card assembly. A circuit card assembly is a grouping of two or more physically connected or related electrical and/or electronic parts capable of disassembly. Each component of the assembly must be capable of functioning in accordance with its own item name. Consists of a printed-wiring board upon which are mounted separately manufactured electronic components, such as capacitors, inductors, resistors, and the like. It may also include printed electronic components. (Source: H6)

3.38 Circuit hole. See "Terminal hole."

3.39 Circumferential separation. A circumferential separation is a circular crack around the barrel of a plated-through hole or eyelet.

3.40 Clad or cladding. Clad or cladding is a relatively thin layer or sheet of metal foil which is bonded to the base material.

3.41 Clearance hole. A clearance hole is a hole in the conductor pattern larger than, but co-axial with, a hole through the printed board. (See figure 4.)

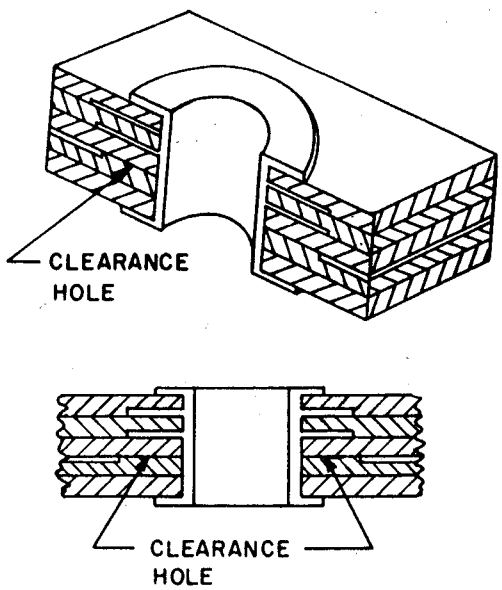


FIGURE 4. Clearance hole.

3.42 Clearance-hole technique. Clearance-hole technique is a fabricati· ·echnique for producing multilayer printed boards wherein all terminal areas on each o· the conductor layers are exposed directly to dip soldering in assembly of components. (See also "Access hole.")

3.43 Clinched leads. Clinched leads are component lead wires which extend through the printed board and are bent, or clinched, so as to make mechanical and electrical contact with the circuitry, for the purpose of improving electrical reliability.

3.44 Component. A component is a separate part of a printed-circuit assembly or printed-wiring assembly which performs a circuit function (e.g. resistor, capacitor, transistor, transformer, etc.).

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3.45 Component mounting. Component mounting is the act of mechanically attaching the component to the printed board, or the manner in which they are attached, or both.

3.46 Component orientation. Component orientation is the direction in which the components, on a printed board or other assembly, are lined-up electrically with respect to the polarity of polarized components and also with respect to one another and to the board.

3.47 Component side. The component side is the side of a printed board upon which components are mounted.

3.48 Composite board. A composite board is a completely laminated multilayer printed board.

3.49 Conductive foil. See "Foil".

3.50 Conductor. A conductor is a single conductive line forming an electrical connection between terminal areas. (In printed board terminology, the terms "conductor", "conductor line", "conductor path", "line", "path", "trace", and "track" are synonymous; but the preferred term is "conductor".)

3.51 Conductor layer. A conductor layer is a layer of a printed board containing conductors. It also includes both ground planes and voltage planes.

3.52 Conductor layer no. 1. Conductor layer no. 1 is the conductor layer of a multilayer printed board on the component side.

3.53 Conductor pattern. A conductor pattern is a pattern of conductors in a conductor layer.

3.54 Conductor side. The conductor side is the side of a single-sided printed board containing the conductors.

3.55 Conductor spacing. Conductor spacing is the distance between adjacent edges of two adjacent conductors on any single conductor layer of a printed board. (See figure 5.)

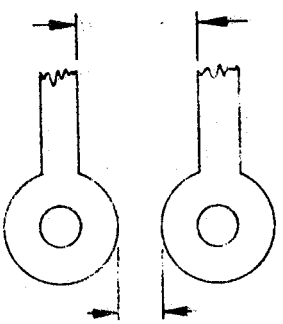


FIGURE 5. Conductor spacing.

3.56 Conductor thickness. Conductor thickness is the thickness of the conductor, exclusive of coatings or other metals.

3.57 Conductor width. Conductor width is the observable width of the conductor at any point chosen at random on the printed board viewed from vertically above. Imperfections, for example nicks, pinholes, or scratches, allowable by the relevant specification shall be ignored.

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3.58 Conformal coating. Conformal coating is an insulating protective coating, which conforms to the configuration of the object coated, applied on the completed board assembly.

3.59 Connection. A connection is the means by which electrical contact is made to the conductor pattern.

3.60 Connector keying pin. Connector keying pin is a pin which permits a plug to be plugged into its mating receptacle; but prevents it from being plugged into any other receptacle. (See also "Polarizing pin.")

3.61 Connector, printed-circuit, two-part. A two-part, printed-circuit connector consists of two plastic bodies containing the pin and socket electrical contacts, one of which mounts on the printed board (as a separate hardware item) and the other which mounts on a chassis or other hardware and accepts the first for making electrical contact. There are two major categories, as follows:

(a) Hook-up-wiring receptacle.
(b) Printed-board plug or receptacle.

3.62 Connector receptacle, printed-board, one-part. A one-part, printed-board, connector receptacle is a connector receptacle containing contacts which accept the edge of the printed board upon which the printed wiring serves as the mating contacts (see "Printed contact"). A receptacle of this type is also called a "card-edge connector", "card-insertion connector", "one-part printed circuit connector", or "one-piece printed-circuit connector."

3.63 Control drawing. Use "Master drawing."

3.64 Coupon. A coupon is a specimen of a printed board, or of printed-board material, for testing purposes. (See also "Quality-conformance-test coupon".)

3.65 Cover layer. The cover layer is the outer layer or layers of insulating material over the conductors.

3.66 Crazing (base material). Crazing is an internal condition occuring in the laminated base material in the form of connected white spots or crosses (measling). It is usually made visible by mechanical or thermal stresses or both. This connected measling forms continuous paths which may carry moisture or electric current or both.

3.67 Crazing (conformal coating). Crazing is a network of fine cracks, extending on or under the surface of, or through a layer of the conformal coating.

3.68 Critical defect. A critical defect is a defect that judgment and experience indicate is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product; or a defect that judgment and experience indicate is likely to prevent performance of the tactical function of a major end item such as a ship, aircraft, tank, missile, or space vehicle. (Source: MIL-STD-105)

3.69 Cross hatching. Cross hatching is the breaking up of large conductive areas by use of a pattern of voids in the conductive material.

3.70 Current-carrying capacity. Current-carrying capacity is the maximum current which can be continuously carried, under specified conditions by an individual conductor, without causing objectionable degradation of electrical or mechanical properties of the printed board.

3.71 Datum points. Datum points are points, assumed to be exact, for purposes of dimensional computation from which the location or geometric relationship of a printed-board feature may be established.

3.72 Defect. A defect is any nonconformance of the unit of product with specified requirements. (Source: MIL-STD-105)

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3.73 Definition. Definition is the fidelity of reproduction of the pattern edge in the printed board (conductors, inductors, etc.) relative to the original master pattern.

3.74 Delamination. Delamination is a separation between plies within the base laminate, between any of the layers of the base laminate and B-stage material, or between the laminate and the metal cladding.

3.75 Dents. Dents are depressions in the foil which do not significantly decrease its thickness.

3.76 Dewetting. Dewetting is a condition which results when the molten solder has coated the surface and then receded leaving irregularly shaped mounds of solder separated by areas covered with a thin solder film; basis metal not exposed. (See figure 6.)

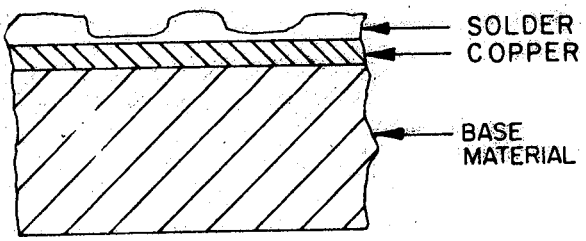


FIGURE 6. Dewetting.

3.77 Dielectric strength. Dielectric strength is the capability of an insulator to withstand an applied voltage without breaking down and conducting current.

3.78 Dimensional stability. Dimensional stability is the degree of freedom from distortion caused by such factors as temperature changes, humidity changes, age, and handling.

3.79 Dip soldering. Dip soldering is a process whereby printed-circuit assemblies or printed-wiring assemblies are brought in contact with the surface of a static pool of molten solder for the purpose of soldering the entire exposed conductor pattern in one operation.

3.80 Double-sided board. A double-sided board is a printed board with conductive material on both sides of the board.

3.81 Drag soldering. Drag soldering is a process whereby supported, moving printed-circuit assemblies or printed-wiring assemblies are brought in contact with the surface of a static pool of molten solder. The assemblies are preceded by an attachment for dross removal.

3.82 Dross. Dross is oxide and other contaminants which form on the surface of molten solder. Incorrectly referred to as slag.

3.83 Edge board contacts. See "Printed edge board contacts."

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3.84 Edge spacing. Edge spacing is the distance of a pattern, or the parts, or both from the edges of a printed board.

3.85 Edge-to-edge spacing. See "Conductor spacing."

3.86 Electroless deposition. Electroless deposition is a type of metal deposition which employs a chemical reducing agent to reduce a metal salt, and to deposit metal onto a catalyzed surface.

3.87 Embossed-foil printed board. See "Stamped printed board."

3.88 Embossing. Embossing is a process consisting of indenting foil into an insulating base, and mechanically removing the unwanted raised portions of foil.

3.89 Engineering drawing. An engineering drawing is an engineering document that discloses by means of pictorial or textual presentations, or combinations of both, the physical and functional end product requirements of an item. (Source: MIL-STD-100)

3.90 Etchant. An etchant is a solution, used to remove by chemical reaction, the unwanted portion of material from a printed board.

3.91 Etchback. Etchback is the chemical process of removing glass fibers and resin between the conductor layers, in order to facilitate a three-surface mechanical and electrical bond on the protruding portion of the conductor materials in the plated-through holes.

3.92 Etched printed board. An etched printed board is a printed board formed by etching.

3.93 Etch factor. Etch factor is the ratio of depth of etch (conductor thickness) to lateral etch (undercut). (See figure 7.)

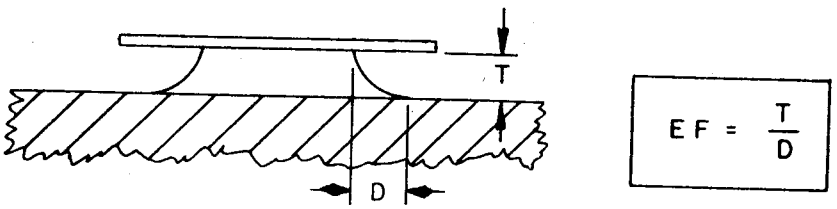


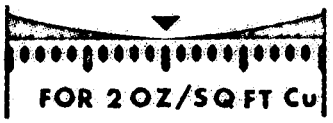
FIGURE 7. Etch factor.

3.94 Etching. Etching is a process wherein a printed pattern is formed by chemical, or chemical and electrolytic, removal of the unwanted portion of conductive material bonded to a base.

3.95 Etching guide. See "Etching indicator."

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3.96 Etching indicator. An etching indicator is a wedge shaped or other specified pattern affixed to the foil to indicate the quality of etching. (See figure 8.)



Etching indicator showing proper etching and correct line width



Etching indicator showing over-etching and insufficient line width

FIGURE 8. Etching indicator.

3.97 External layer. An external layer is a conductor pattern or terminal area on the surface of the composite board.

3.98 Fabrication holes. Use "Indexing holes."

3.99 Failure. Failure is the inability of an item to perform within previously specified limits.

3.100 Feedthrough (printed board). Use "Interfacial connection."

3.101 Fiber exposure. Fiber exposure is a condition in which glass cloth fibers are exposed in machined or abraded areas or in chemically attacked areas. (See also "Weave exposure.")

3.102 Fill. The fill consists of the threads of the reinforcing glass fabric that run crosswise in the basic roll of glass fabric when woven or processed. Sometimes also called woof. The resulting laminated base material is usually not as strong in the fill direction as in the warp direction.

3.103 Finger. Use "Printed contact."

3.104 First article. The first article is a sample part or assembly manufactured, after award of contract, but prior to the start of production, for the purpose of assuring the buyer that the manufacturer is capable of manufacturing a product which will meet the requirements.

3.105 First article inspection. First article inspection is inspection that is performed, after award of contract but prior to the start of production, for the purpose of assuring the buyer that the manufacturer is capable of manufacturing a product which will meet the requirements.

3.106 Flexible printed wiring. Flexible printed wiring is multiconductor, etched, flexible, flat sealed wiring, generally custom designed for a specific application.

3.107 Flush conductor. A flush conductor is a conductor whose outer surface is in the same plane as the surface of the insulating base.

3.108 Flush printed board. A flush printed board is a printed board in which the outer surface of the reproduced pattern is in the same plane as the outer surface of the insulating base.

3.109 Flux. Flux is a chemically active compound that is capable of promoting the wetting of metals with solder.

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3.110 Fluxing. Fluxing is the act of applying flux to the conductor and terminal areas, prior to soldering.

3.111 Foil. Foil is a very thin sheet of metal.

3.112 Fusing. Fusing is the melting of a (usually electrodeposited) metallic coating, followed by solidification.

3.113 Graded wedge. See "Etching indicator."

3.114 Grid. A grid is an orthogonal network consisting of equidistant lines providing the basis for an incremental location system.

3.115 Ground plane. A ground plane is a continuous sheet of metal on or in a printed board, used as a common reference point for circuit returns, shielding, and heat sinking.

3.116 Ground-plane clearance. Ground-plane clearance is the etched portion of a common ground plane around a plated-through or nonplated-through hole which prevents connection between the ground plane and the hole. (See figure 9.)

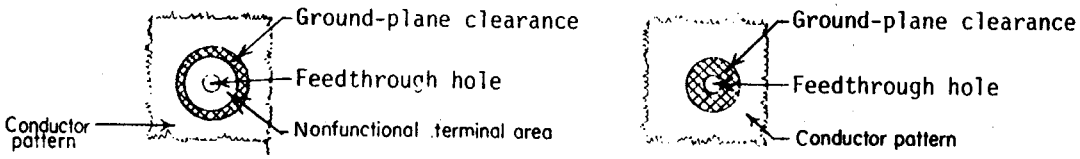


FIGURE 9. Ground-plane clearance.

3.117 Haloing. Haloing is a form of delamination or measling, or both, caused by mechanical stress and usually occurs around punched or drilled holes. It may not become visible until after exposure to thermal stress.

3.118 Hard wiring. Hard wiring is electrical wiring that interconnects two or more parts or assemblies into an assembly which is inseparable without the use of special tools and techniques.

3.119 Hole. See also "Access hole", "Circuit hole", "Clearance hole", "Fabrication holes", "Indexing holes", "Interstitial via hole", "Lead mounting holes", "Mounting hole", "Pilot holes", "Pinholes", "Plated-through hole", "Reference hole", "Supported hole", "Terminal area clearance hole", "Terminal hole", "Tooling holes", "Unsupported hole", and "Via hole".

3.120 Hole-pull strength. Hole-pull strength is the mechanical strength a plated-through hole or terminal area exhibits when it is loaded or pulled in the direction of the axis of the hole.

3.121 Hook-up wiring receptacle. A hook-up wiring receptacle is a receptacle where the terminations are designed for the attachment of one or more hook-up wires.

3.122 Icicle. Use "Solder projection."

3.123 Inclusions. Inclusions are foreign particles, metallic or nonmetallic, which may be entrapped in the laminate or foil and could form a conductive path between conductors.

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3.124 Indexing holes. Indexing holes are holes placed in a printed-board blank to enable accurate positioning for processing. (Indexing holes may or may not be on the finished board.)

3.125 Indexing notch. An indexing notch is a notch placed in the edge of a printed board blank to enable accurate positioning for processing. (Indexing notches may or may not be on the finished board.)

3.126 Inspection. An inspection is a process of measuring, examining, testing, or otherwise comparing the unit of product with the requirements.

3.127 Inspection lot. An inspection lot is a collection of units of product bearing identification and treated as a unique entity from which a sample is to be drawn and inspected to determine conformance with the acceptability criteria.

3.128 Insulation resistance. Insulation resistance is the electrical resistance of the insulating material between any pair of contacts, conductors, or grounding devices in various combinations.

3.129 Interconnection. Interconnection is the physical wiring between components (outside a module), between modules, between units, or between larger portions of a system or systems.

3.130 Interfacial connection. An interfacial connection is a conductor which connects conductor patterns on opposite sides of the base.

3.131 Interlayer connection. An interlayer connection is a conductor which provides electrical continuity to or between layers.

3.132 Internal layer. An internal layer is a conductor pattern which is contained entirely within the composite board.

3.133 Interstitial via hole. See "Via hole."

3.134 Jumper wire. A jumper wire is a direct electrical connection, which is not a portion of the conductor pattern, between two points on a printed board, added after the printed board is fabricated.

3.135 Key. See "Register mark."

3.136 Keying pin. See "Connector keying pin."

3.137 Keying slot. A keying slot is a slot in a printed board which permits the printed board to be plugged into its mating receptacle; but prevents it from being plugged into any other receptacle. (See also "Polarizing slot.")

3.138 Laminate. Laminate is the product made by bonding together two or more layers of material with heat, pressure, and an adhesive.

3.139 Laminate void. A laminate void is an absence of resin in any cross section area which should normally contain resin.

3.140 Laminating. Laminating is the process of bonding two or more layers of material together, using heat, pressure, and an adhesive (see B-stage).

3.141 Land. Use "Terminal area."

3.142 Layer-to-layer registration. Layer-to-layer registration is the relative position of two or more conductor patterns with respect to their desired locations on or within a multilayer printed board.

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3.143 Layer-to-layer spacing. Layer-to-layer spacing is the distance between adjacent surfaces of conductor patterns in adjacent layers. (See figure 10.)

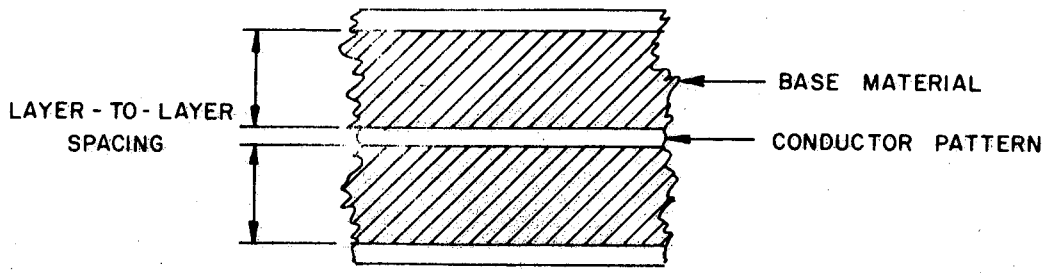


FIGURE 10. Layer-to-layer spacing.

3.144 Lead mounting holes. Lead mounting holes are the holes in a printed board through which component leads pass.

3.145 Legend. Legend is letters, numerals, or symbols on the printed board indicating part numbers, component locations, etc.

3.146 Major defect. A major defect is a defect, other than critical, that is likely to result in failure, or to reduce materially the usability of the unit of product for its intended purpose. (Source: MIL-STD-105)

3.147 Manufacturing holes. Use "Indexing holes."

3.148 Marking. Marking is a method of identifying printed boards with information, such as part number, revision letter, manufacturer's trade mark, etc.

3.149 Mask. A mask is a material applied to enable selective etching, plating, or application of solder to a printed board. A mask may be a photochemical material applied as a sheet, liquid, etc., or a nonphotochemical material which is applied by hand, silk screen, etc.

3.150 Master drawing. A master drawing is a drawing showing the dimensional limits or grid location applicable to any or all parts of a printed board, including the base.

3.151 Master pattern. A master pattern is a 1-to-1 scale pattern which is used to produce the printed board within the accuracy specified in the master drawing.

3.152 Mealing. Mealing is a condition at the interface of the conformal coating and base material, in the form of discrete spots or patches, which reveals a separation of the conformal coating from the surface of the printed board, of from the surfaces of attached components, or from both.

3.153 Measling. Measling is an internal condition occurring in the laminated base material in which the glass fibers are separated from the resin. This condition exists in the form of discrete white spots or crosses evident at individual and scattered weave intersections. It is usually made visible by mechanical or thermal stresses or both.

3.154 Metal-clad base material. Metal-clad base material is a material which consists of an insulating base with metal foil bonded to one or both surfaces.

3.155 Microstrip. A microstrip is a type of transmission line configuration which consists of a conductor over a parallel ground plane, and separated by a dielectric.

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3.156 Minor defect. A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose, or is a departure from established standards having little bearing on the effective use or operation of the unit. (Source: MIL-STD-105)

3.157 Mother board. A mother board is a printed-wiring assembly which interconnects arrays of plug-in printed-circuit assemblies or printed-wiring assemblies. Also known as a platter. (See also "Backpanel.")

3.158 Mounting hole. A mounting hole is a hole used for mechanical mounting of a printed board to the chassis or for mechanical attachment of components to the board.

3.159 Multilayer printed board. Multilayer printed board is a general term for either multilayer printed circuit or multilayer printed wiring. It includes flexible or rigid multilayer boards containing more than two conductive layers.

3.160 Multilayer printed circuit. A multilayer printed circuit consists of alternate layers of conductor circuitry and insulating materials bonded together, with conductive layers interconnected as may be required.

3.161 Multilayer printed wiring. Multilayer printed wiring consists of alternate layers of conductors and insulating material bonded together, suitable for use as an assembly and component support and for electrically interconnecting components.

3.162 Nailheading. Nailheading is a flared condition of internal conductors caused by improper drilling. (See figure 11.)

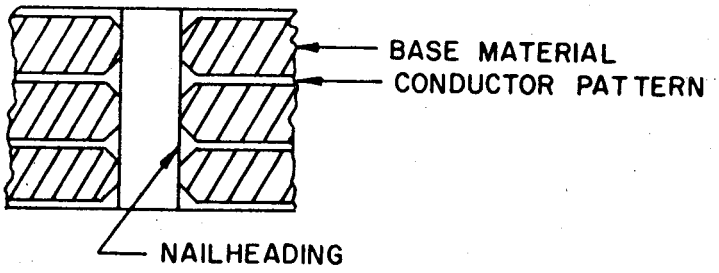


FIGURE 11. Nailheading.

3.163 Negative, artwork. The artwork negative is the original production master or the production master in which the conductor pattern is transparent, with an opaque background.

3.164 Nonconductive pattern. A nonconductive pattern is a design formed by functional nonconductive material of a printed board.

3.165 Nonflammable. Nonflammable is that property of a material which causes it to be incapable of bursting into flame when a spark or open flame is passed sufficiently near. A nonflammable material is not necessarily noncombustible.

3.166 Nonfunctional terminal area. A nonfunctional terminal area is a terminal area that does not have any connecting conductors and which may be located on internal or external layers.

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3.167 Nonwetting. Nonwetting is a condition whereby a surface has contacted molten solder, but the solder has not adhered to all of the surface. (See figure 12.)

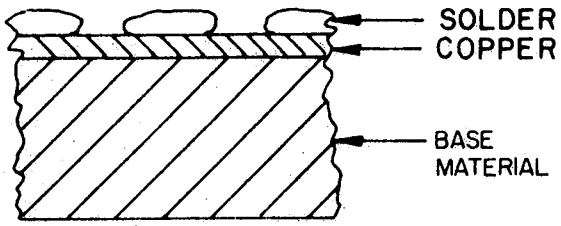


FIGURE 12. Nonwetting.

3.168 Original drawing. An original of a drawing is the drawing or "Marked" copy thereof on which is kept the revision record recognized as official by the design activity. (Source: MIL-STD-100)

3.169 Overhang. Overhang is the increase in conductor width, caused by plating build-up. It is measured from one side of a conductor. (See figure 13.)

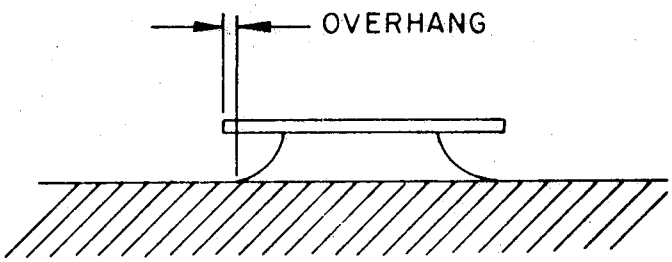


FIGURE 13. Overhang.

3.170 Overplate. Overplate is the outer layer of plated metal.

3.171 Pad. See "Terminal area."

3.172 Panel. A panel is a rectangle or square of predetermined size containing one or more printed boards and quality-conformance-test coupons, if required.

3.173 Panel plating. Panel plating is plating of the entire surface (including holes) of a panel.

3.174 Part. A part is one piece, or two or more pieces joined together, which are not normally subject to disassembly without destruction of designed use. (Source: MIL-STD-100)

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3.175 Pattern plating. Pattern plating is the plating of the printed-circuitry pattern, including hole walls.

3.176 Peel strength. Peel strength is a measure, in pounds per inch width, of the force required to separate a strip of conductive material from the adjoining base by a pull applied to the edge of the conductive material.

3.177 Photomask. A photomask is a mask consisting of photo-resist material, and applied by the photomasking process.

3.178 Photo resist. A photo resist is a photosensitive, chemically resistant polymeric material which is used for photomasking of printed boards.

3.179 Pilot holes. Use "Indexing holes."

3.180 Pinholes. Pinholes are small holes occurring as imperfections which penetrate entirely through a material.

3.181 Pits. Pits are small holes occurring as imperfections which do not penetrate entirely through the foil.

3.182 Plated printed wiring. Plated printed wiring is printed wiring which is plated with a metal over the printed wiring.

3.183 Plated resist. Plated resist is a metallic plating used to protect certain portions of the printed conductor pattern from the action of the etchant.

3.184 Plated-through hole. A plated-through hole is an interfacial connection formed by the deposition of conductive material on the sides of the hole through the base material.

3.185 Plating. Plating is the process consisting of the chemical or electrochemical deposition of metal.

3.186 Plating up. Plating up is a process for depositing additional metal or metals onto a conductor pattern.

3.187 Plating void. Plating void is the area of absence of a specific metal from a specific cross sectional area.

3.188 Polarizing pin. A polarizing pin is a connector pin that has been made so that it may be inserted into its mating receptacle in only one position. (See also "Connector keying pin.")

3.189 Polarizing slot. Polarizing slot is the slot, in the edge of a printed board, used to assure proper insertion and location in a mating connector. (See also "Keying slot.")

3.190 Positive pattern. The positive pattern is the image on the production master in which the conductor pattern is opaque on a transparent background.

3.191 Post. A post is a solid plating build-up for interlayer connection.

3.192 Preheating. Preheating is a heating operation used to raise the temperature of the work above room temperature and thus reduce the thermal shock of an elevated temperature processing operation.

3.193 Preimpregnated bonding sheet. Use "Prepreg."

3.194 Prepreg. Prepreg is sheet material, usually fabric or paper, which has been impregnated with a thermosetting resin and partially cured to a B-stage.

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3.195 Pretinning. See "Tinning."

3.196 Printed board. Printed board is the general term for either printed circuits or printed-wiring board. It includes flexible or rigid, single-sided, double-sided, or multilayer boards. (See also "Multilayer printed board.")

3.197 Printed-board assembly. A printed-board assembly is an assembly of several printed-circuit assemblies or printed-wiring assemblies, or both.

3.198 Printed-board plug or receptacle. A printed-board plug or receptacle is a plug or receptacle where the contact terminations are designed for direct mounting on a printed board.

3.199 Printed circuit. A printed circuit is a pattern composed of printed wiring and printed components all formed on a common base.

3.200 Printed-circuit assembly. A printed-circuit assembly is a board having printed components and on which separately manufactured components and parts have been added.

3.201 Printed-circuit board. A printed-circuit board is a completely processed conductor pattern or patterns, and printed-components all formed on a common base.

3.202 Printed component. A printed component is a component such as a printed inductor, resistor, capacitor, transmission line, etc., which is in printed form.

3.203 Printed contact. A printed contact is that portion of printed wiring used for the purpose of providing electrical connection.

3.204 Printed edge board contacts. Printed edge board contacts are patterns extending to the edge of a printed board intended for mating with external contacts.

3.205 Printed wiring. Printed wiring is a conductor pattern used for the purpose of providing point-to-point electrical connections.

3.206 Printed-wiring assembly. A printed-wiring assembly is a printed-wiring board on which separately manufactured componenets and parts have been added.

3.207 Printed-wiring board. A printed-wiring board is a completely processed conductor pattern or patterns, all formed on a common base.

3.208 Printing. Printing is the act or art of reproducing a pattern on a surface.

3.209 Procuring activity. The procuring activity is the organizational element of the Government which contracts for articles, supplies, or services, or it may be a contractor or subcontractor when the organizational element of the Government has given specific written authorization to such contractor or subcontractor to serve as agent for the procuring activity. A contractor or subcontractor serving as agent for the procuring activity shall not have the authority to grant waivers, deviations, or exceptions to the contract unless specific written authorization to do so has also been given by the Government organizational element.

3.210 Production board. A production board is a finished board that the contractor is supplying in production.

3.211 Pull strength. Use "Bond strength." (See also "Hole-pull strength.")

3.212 Quality-conformance-test coupon. A quality-conformance-test coupon is a test coupon made as an integral part of the printed-board panel, on which specified tests are performed for evaluation without destroying the basic board.

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3.213 Reference dimension. A reference dimension is a dimension without tolerance used only for information purposes that does not govern production or inspection operations.

3.214 Reference grid. Use "Grid."

3.215 Reference hole or notch. Use "Indexing hole" or "Indexing notch," as applicable.

3.216 Reflow. Use "Fusing."

3.217 Reflow soldering. Reflow soldering is a process for joining parts by tinning the faying surfaces, placing them together, heating until the solder fuses, and allowing to cool in the joined position.

3.218 Register. Register is the relative position of one or more printed patterns, or portions thereof, with respect to their desired locations on the base material.

3.219 Register mark. A register mark is a mark used to establish the registration. (See figure 14.)

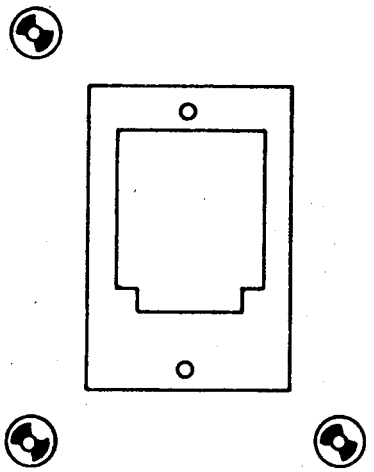


FIGURE 14. Register mark.

3.220 Registration. Registration is the degree of register.

3.221 Repairing. Repairing is the act of restoring the functional capability of a defective part, without necessarily restoring appearance and uniformity.

3.222 Resin smear. Resin smear is resin which has been deposited on edges of copper in holes during drilling, either as uniform coating or scattered patches. Resin smear electrically isolates the conductor layers from the plated-through hole interconnections.

3.223 Resist. Resist is a material (ink, paint, metallic plating, etc.) used to protect the desired portions of the printed pattern from the action of the etchant, solder, or plating.

3.224 Reversion. Reversion is a chemical reaction in which a polymerized material degenerates at least partially, to a lower polymeric state or the original monomer. It is usually accompanied by significant changes in physical and mechanical properties.

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3.225 Reworking. Reworking is the act of repeating one or more manufacturing operations or performing alternate operations for the purpose of improving the yield of acceptable parts.

3.226 Screen process printing. Screen process printing is a process for affixing an image onto base material, by forcing ink through a stencil screen with a squeegee.

3.227 Seeding. Seeding is a sensitizing process for catalytically initiating metal deposition by chemical reduction.

3.228 Self-extinguishing. Self-extinguishing is that property of a material which causes it to stop burning if the heat source is removed.

3.229 Semiadditive process. The semiadditive process is the process for obtaining conductor patterns which combines electroless metal deposition with electro-plating or etching to produce patterns.

3.230 Sensitizing. See "Seeding."

3.231 Sequential laminating process. The sequential laminating process is a process for making multilayer printed boards by laminating increments of double-sided boards together. The circuitry layers are interconnected with via holes and through connections.

3.232 Separable component. A separable component is a replaceable component, the body of which is not chemically bonded to the base, excluding the effects of protective coatings, solder, and potting materials.

3.233 Shadowing. Shadowing is an etchback defect in which the dielectric material immediately next to the foil is inadequately removed although maximum etchback has been achieved elsewhere. (See figure 15.)

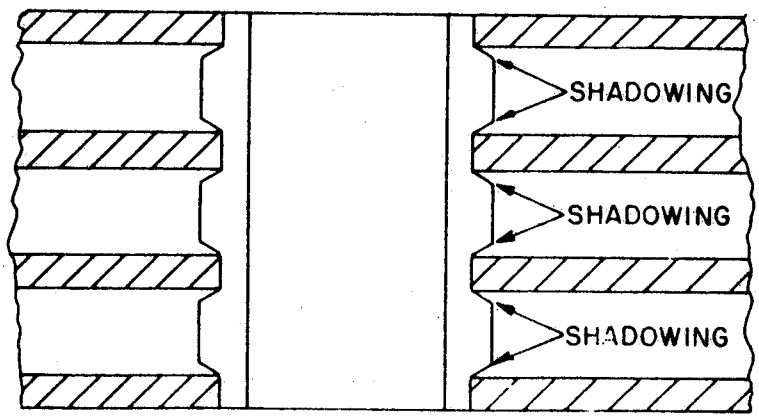


FIGURE 15. Shadowing.

3.234 Single-sided board. A single-sided board is a printed board with printed conductors on one side.

3.235 Smear removal. See "Chemical hole cleaning."

3.236 Solderability. Solderability is the property of a metal to be wetted by solder.

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3.237 Solderless wrapped connection. A solderless wrapped connection is a solderless connection made by wrapping several turns of bare wire under tension around a sharp-corner rectangular terminal, using either a power tool or hand tool.

3.238 Solder projection. A solder projection is an undesirable protrusion from a solidified solder joint or coating.

3.239 Solder resist. See "Resist". NOTE: The term "solder resist" has several different meanings depending on usage.

3.240 Spacing. See also "Center-to-center spacing", "Conductor spacing", "Edge spacing", "Edge-to-edge spacing", and "Layer-to-layer spacing."

3.241 Specimen. A specimen is a fraction or the whole of a sample unit, or a representative example of material evolved from the identical process as the product.

3.242 Stable base material. Stable base material is a drafting material, having good dimensional stability, used for making artwork and master patterns.

3.243 Stamped printed board. A stamped printed board is a printed board formed on an insulating base by die stamping.

3.244 Substractive process. A substractive process is a process for obtaining conductor patterns, consisting of selective removal of the unwanted portions of the foil.

3.245 Supported hole. A supported hole is a hole in a printed board that has its inside surface plated or otherwise reinforced.

3.246 Surface mounting. Surface mounting is the electrical connection of components on the surface of a conductor pattern without utilizing lead mounting holes.

3.247 Swaged leads. Swaged leads are component lead wires which extend through the printed board and are flattened, or swaged, so as to secure the component to the board during manufacturing operations.

3.248 Tab. See "Printed contact." (See also "Printed edge board contacts.")

3.249 Terminal area. A terminal area is a portion of a printed board used for making electrical connections to the conductor pattern, such as the enlarged portion of conductor material surrounding a lead mounting hole.

3.250 Terminal area clearance hole. See "Access hole."

3.251 Terminal hole. A terminal hole is a hole within a terminal area.

3.252 Terminal pad. Use "Terminal area."

3.253 Test coupon. See "Coupon."

3.254 Test pattern. A test pattern is a pattern affixed to a printed board or a quality-conformance-test coupon for inspection or testing purposes.

3.255 Test specimen. A test specimen is a printed board, selected from a batch of boards, for testing. Test specimens are manufactured simultaneously together with other boards, and are representative of the quality of the entire batch.

3.256 Through connection. Use "Interfacial connection."

3.257 Tinning. Tinning is a hot dip process for the application of solder coating on conductors and terminals, primarily for minimizing in-process oxidation and enhancing solderability.

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3.258 Tooling holes. Tooling holes are holes placed in a printed board to enable the base to be positioned accurately for processing.

3.259 Total board thickness. See "Board thickness."

3.260 Trace. Use "Conductor."

3.261 Transferred printed board. A transferred printed board is a printed board in which the pattern is formed on a temporary base and transferred to the permanent base.

3.262 True position diameter. The true position diameter is the diameter of the circular or cylindrical tolerance zone which surrounds a true position location.

3.263 True position tolerance. The true position tolerance is the total diameter of permissible movement around the true position as shown in the master drawing.

3.264 Twist. Twist is a bending or curving distortion from a true or plane surface, in a direction parallel to a diagonal between two opposite corners of the sheet or board. (See also "Warp (base material).") (See figure 16.)

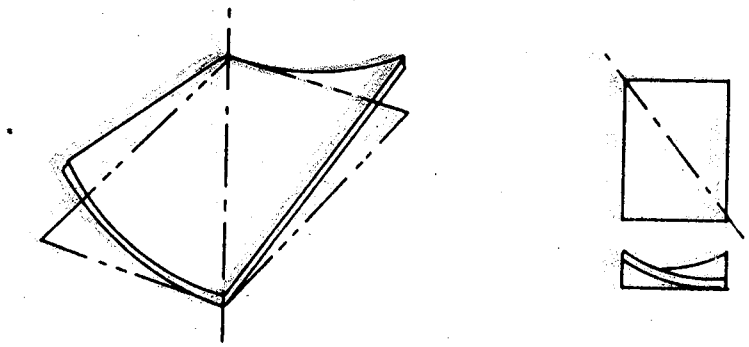


FIGURE 16. Twist.

3.265 Undercut. Undercut is the reduction of the cross section of a conductor caused by the etchant removing metal from under the edge of the resist or plating. (See figure 17.)

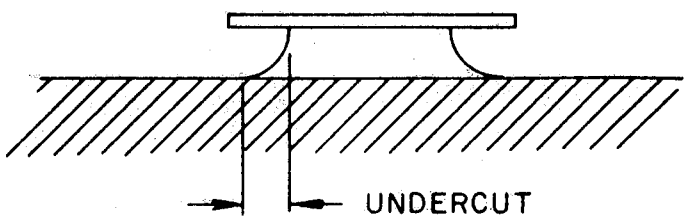


FIGURE 17. Undercut.

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3.266 Unsupported hole. An unsupported hole is a hole containing no conductive material, plating, solder, or any type of reinforcement.

3.267 Unwetting. Use "Nonwetting."

3.268 Vapor deposition. Vapor deposition is a process consisting of condensation of a material from its gaseous state onto a base.

3.269 Via hole. A via hole is a plated-through hole which serves as a through connection between two or more internal conductor layers within a multilayer printed board.

3.270 Voltage plane. A voltage plane is a continuous sheet of metal on or in a printed board which functions at other than ground potential. It can also be used as a common point for circuit returns, shielding, and heat sinking.

3.271 Voltage-plane clearance. Voltage-plane clearance is the etched portion of a common voltage plane around a plated-through or nonplated-through hole which prevents connection between the voltage plane and the hole.

3.272 Warp (base material). Warp is the deviation from flatness of a board characterized by a roughly cylindrical or spherical curvature such that, if the board is rectangular, its corners or edges are in the same plane as the major surfaces of the board. (See figure 18.)

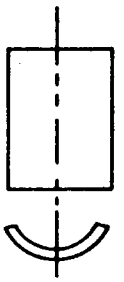


FIGURE 18. Warp (base material).

3.273 Warp (fabric). The warp consists of the threads of the reinforcing glass fabric that run in the machine or rolled direction of the basic roll of glass fabric when woven or processed. The resulting laminated base material is usually stronger in the warp direction than in the fill direction.

3.274 Wave soldering. Wave soldering is a process whereby printed boards are brought in contact with the surface of continuously flowing and circulating solder.

3.275 Weave exposure. Weave exposure is a surface condition in which the glass cloth is exposed (i.e. locally devoid of resin). It usually occurs because of chemical attack or mechanical abrasion. (See also "Fiber exposure.")

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3.276 Wetting. Wetting is the free flow and spreading of solder on a metallic surface to form an adherent bond. (See figure 19.)

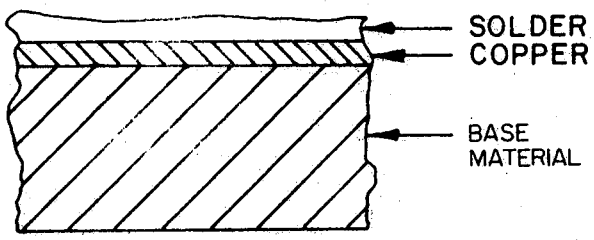


FIGURE 19. Wetting.

3.277 Whisker. A whisker is a slender acicular metallic growth which usually occurs after the printed board has been manufactured.

3.278 Wicking (chemical). Chemical wicking is the migration of chemicals into the glass fibers of the insulating material.

3.279 Wicking (solder). Solder wicking is the action of drawing up solder through capillary action.

3.280 Wiring pattern. Use "Conductor pattern."

4. GENERAL REQUIREMENTS.

4.1 Not applicable.

5. DETAIL REQUIREMENTS.

5.1 Not applicable.

Custodians:
Army - EL
Navy - EC
Air Force - 17

Review activities:
Army - MI
Navy - AS, MC, SH
Air Force - 11, 26, 80, 85
DSA - ES, GS
NS - S23

User activities:
Army - ME, MU, PA, WC
Navy - CG, OS
Air Force -

Preparing activity:
Air Force - 17

Agent:
DSA - ES

(Project 5999-0060)

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